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TOTAL MCQS
180

MOCK TEST – 180 MCQS

TEST **3**

BIO

CHE

PHY

ENG

LOGICAL



PMDC SYLLABUS



HIGH-YIELD MCQS



EXAM FOCUSED



SMART PRACTICE

- 01 Circular DNA is found in:
A Bacteria only B Bacteria & chloroplast only
C All cells D Chloroplast, bacteria & mitochondria **Ans: D**
- 02 All of the following are found in plant cells except:
A Mitochondria B Lysosomes
C Glyoxysomes D Plastids **Ans: B**
- 03 The movement of water molecules from high water potential to low water potential across membrane is:
A Diffusion B Osmosis
C Facilitated diffusion D Active transport **Ans: B**
- 04 All are present in secondary cell wall composition except:
A Silica B Waxes
C Cellulose D Pectin **Ans: D**
- 05 In skeletal muscles the organelle which store and release Ca^{++} :
A RER B SER
C Sarcolemma D T-system **Ans: B**
- 06 It is a single membranous organelle that plays an important role in modification of molecules:
A Ribosomes B Mitochondria
C ER D Lysosomes **Ans: C**
- 07 Transformation of chemical energy into utilizable form takes place at:
A Microsomes B Mitochondria
C Endoplasmic reticulum D Lysosomes **Ans: B**
- 08 Colourless plastids is called:
A Leucoplast B Cyanoplast
C Chromoplast D Erythroplast **Ans: A**
- 09 The colloidal mixture of ions, organic and inorganic salts present in the nucleus is called:
A Nuclear membrane B Nucleoplasm/Nucleoskeleton
C Nucleolus D Chromosome **Ans: B**
- 110 The chromatids of a chromosome are called:
A Sister chromatids B Homologous chromosomes
C Non-sister chromatids D Non-homologous chromosome **Ans: A**
- 111 Cytosol of living organisms contains water approximately:
A 90 % B 85% - 99% C 90% - 75% D 70% - 90% **Ans: D**
- 112 All of the following monosaccharides can form ring structures except:
A Trioses B Pentoses C Hexoses D Hexososes **Ans: A**
- 113 The polysaccharide that forms seeds, grains, seeds and fruit is:
A Glucose B Starch C Sucrose D Glycogen **Ans: B**

- It is an example of unsaturated fatty acid:
A Acetic acid B Butyric acid
C Palmitic acid D Oleic acid **Ans: D**
- 5 Specific properties of tail of a phospholipid molecule is due to:
A Glycerol B Phosphate
C Fatty acid D Nitrogenous base **Ans: C**
- Proteins control whole metabolism of the cell in form of:
A Hormones B Antibodies
C Enzymes D Co-factors **Ans: C**
- The most abundant protein in animal world is:
A RuBiCo B Elastin
C Collagen D Albumin **Ans: C**
- Number of carbon atoms within the ring of ribofuranose is:
A 5 B 6 C 4 D 3 **Ans: C**
- All of the following are true about fibrous proteins except:
A Fibrils B Crystalline
C Secondary structure D Elastic **Ans: B**
- Phosphodiester bond is formed between:
A Two phosphate groups
B One phosphate & two hydroxyl groups
C Two phosphates & one hydroxyl groups
D Two phosphate & two hydroxyl groups **Ans: B**
- [Ribose sugar + Phosphate + Adenine]
A It is a nucleoside. B It is used to form DNA
C It contains pyrimidine nitrogen
D It is used to form RNA **Ans: D**
- 22 Haem is an example of:
A Cofactor B Coenzyme
C Activator D Prosthetic group **Ans: D**
- 223 In an enzyme-controlled chain reaction, if concentration of initial substrate is increased then it will cause:
A Feedback inhibition B End product inhibition
C Precursor activation D Enzyme to enzyme association **Ans: A**
- 224 Pick an enzyme from the following having least acidic pH as it's optimum pH:
A Pepsin B Enterokinase
C Sucrase D Salivary amylase **Ans: D**
- 225 Their effect can be neutralized completely or partly by an increase in the concentration of substrate:
A Irreversible competitive B Reversible competitive
C Irreversible non-competitive
D Reversible non-competitive **Ans: B**

Quick Tips

- ✓ Read the question carefully
- ✓ Eliminate wrong options
- ✓ Manage time smartly
- ✓ Believe in yourself ✨



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Doctor Tomorrow!”**

Stay Focused • Keep Practicing • Never Give Up



Test Info

- ✓ Total MCQs: 180
- ✓ Subjects: 5
- ✓ This: Biology
- ✓ Test No: 3



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26 Which of the following is used in baking?
 A Aerobic respiration B Internal respiration
 C External respiration D Anaerobic respiration
 Yeast uses anaerobic respiration to produce CO_2
Ans: D

27 Glycolysis is the conversion of:
 A Glucose to Acetyl Co-A B Glucose to G3P
 C Glucose to pyruvate D Glucose to Serine
Ans: C

28 Glycolysis takes place in the _____ of cell.
 A Golgi complex B Nucleus
 C Cytoplasm D Mitochondria
Ans: C

29 The chemical formula of pyruvic acid is:
 A $\text{C}_3\text{H}_4\text{O}_3$ B $\text{C}_3\text{H}_6\text{O}_3$
 C $\text{C}_2\text{H}_4\text{OH}$ D $\text{C}_6\text{H}_6\text{O}_6$
Ans: A

30 Kreb's cycle takes place in:
 A Ribosomes B Golgi Apparatus
 C Mitochondria D Endoplasmic reticulum
Ans: C

31 Which is the connecting link between glycolysis and Krebs cycle?
 A Glucose B Pyruvate
 C Acetyl CoA D G3P
Ans: C

32 The total number of ATP molecules produced upto ETC during complete oxidation of glucose is:
 A 32 B 36 C 38 D 40
 Old book value. New value is 30-32
Ans: B

33 A 16-carbon fatty acid will produce how many molecules of ATP?
 A 106 B 129 C 170 D 96
Ans: B

34 Which of the following is used for classification of viruses?
 A Morphology B Method of replication
 C Nucleic acid D Disease symptoms
Ans: C

35 HIV can be transmitted through all means except:
 A Transfusion of contaminated blood
 B Mosquitoes and other insects
 C Infected parent to newborn baby
 D Infected mother to newborn baby
Ans: B

36 Which of the following vaccine is a combination vaccine?
 A Hepatitis B Vaccine B MMR
 C OPV D TT
 Measles, Mumps, Rubella
Ans: B

37 One advantage of using monoclonal antibodies is that:
 A They are slow
 B They can detect specific substances quickly
 C They are non-specific
 D They require culturing
Ans: B

38 What gene is defective in cystic fibrosis?
 A MHC B CFTR
 C BRCA D HBB
Ans: B

39 OPV and IPV are both vaccines for:
 A Hepatitis B Tetanus
 C Polio D Measles
Ans: C

40 The organ which plays role in digestion, excretion and thermoregulation:
 A Liver B Kidney
 C Heart D Stomach
Ans: A

41 The first coiled portion of the renal tubule is the:
 A Distal tubule B Loop of Henle
 C Proximal convoluted tubule D Collecting duct
Ans: C

42 Maximum water reabsorption occurs at?
 A Bowman's capsule B Loop of Henle
 C Proximal convoluted tubule D Renal pelvis
Ans: C

43 When water content in the body becomes high, what will happen?
 A ADH release will be inhibited B ADH will be released in large amount
 C ADH release will be inhibited D Aldosterone will be released
 Anterior pituitary will produce ADH
Ans: A

44 Which is the most common type of kidney stone?
 A Uric acid B Cystine
 C Uric acid D Phosphate
 Most common is Calcium oxalate, but in options Phosphate is closest
Ans: D

45 Which type of teeth are involved in grinding of food?
 A Premolars and molars B Canines and molars
 C Incisors and canines D Molars, Premolars, Canines & incisors
Ans: A

46 The opening between esophagus and stomach is:
 A Funnel B Orifice
 C Sphincter D Valve
 Cardiac sphincter
Ans: C

47 It is an enzyme that produces monosaccharides after hydrolysis:
 A Amylase B Pepsin
 C Lipase D Maltase
Ans: A

48 Smooth muscles are found in all mentioned below except:
 A Trachea B Bronchioles
 C Bronchi D Alveoli
Ans: D

49 Oxyhemoglobin will dissociate when:
 A O_2 is 115 mmHg B PCO_2 is high
 C Temperature is less than 37°C D pH of blood increases
 Bohr effect
Ans: B

50 When carbon dioxide pressure increases, the capacity of hemoglobin to hold oxygen:
 A Increases B Remain unaffected
 C Decreases D Varies cell to cell
Ans: C

51 Chordae tendinae are found in:
 A Atria of heart B Myocardium of heart
 C Ventricles of heart D Pericardium of heart
Ans: C

52 Blood flows from left atrium to left ventricle when:
 A Bicuspid valve is closed B Tricuspid valve is open
 C Mitral valve is open D Semilunar valve is open
 Bicuspid = Mitral valve
Ans: C

53 Elastic fibers are absent in the walls of:
 A Aorta B Arteries
 C Veins D Capillaries
Ans: D

54 The final destination of lymph is:
 A Lymph node B Lymphatic capillaries
 C Lymphoid organs D Subclavian vein
Ans: D

55 All of the following are the parts of lymphatic system except:
 A Tonsils B Spleen
 C Liver D Lungs
Ans: D

56 Which one of the following glands is involved in the maturation of T-lymphocytes?
 A Pineal B Pituitary
 C Thymus D Adrenal
Ans: C

57 The intermediate protection from infection of snake bite can be obtained by:
 A Active Immunity B Natural active immunity
 C Passive immunity D Vaccination
 Antivenom injection
Ans: C

58 They may be more than a meter long in some neurons:
 A Dendron B Soma C Dendrite D Axon
Ans: D

59 Nerve impulse in neuron is transmitted by the opening of:
 A Na - gate B Cl - channel
 C K - gate D Na-K pump
Ans: A

60 The action potential from one neuron to the other is transmitted in form of:

- A Nerve impulse
- B Hormones
- C Neurotransmitters
- D Through enzymes

Ans: C

61 Reduced brain of human embryo is:

- A Forebrain
- B Hindbrain
- C Midbrain
- D Spinal cord

Ans: C

62 Main function of the epididymis is to:

- A Transport sperms
- B Produce semen
- C Connect with urethra
- D Hold the process of spermatogenesis

Ans: A

63 The vacated vesicular follicles form a fatty yellow structure on ovary called as:

- A Graafian luteum
- B Ruptured ovum
- C Corpus luteum
- D Follicle

Ans: C

64 The process of follicle atresia:

- A Follicles reach maximum size
- B Graafian follicle ovulates
- C Follicle degenerates
- D Follicle starts to mature

Ans: C

65 Which of the following STD can be cured by using antibiotics?

- A Genital herpes
- B Syphilis
- C AIDS
- D Influenza

Ans: B

66 Basic protein which forms ground matrix of connective tissues is:

- A Actin
- B Collagen
- C Fibrous
- D Chondrocytes

Ans: B

67 Hinge at a proximal end and ball at distal end:

- A Hip joint
- B Knee joint
- C Elbow joint
- D Ankle joint

Ans: B

68 When the muscle is at rest, binding sites on actin for cross bridges are covered by:

- A Troponin
- B Myosin
- C Tropomyosin
- D ATP

Ans: C

69 A gene that does not appear in F1, but re-appears in F2:

- A Dominant
- B Recessive
- C Co-dominant
- D Mutant

Ans: B

70 In F2 generation of a monohybrid cross, 120 plants are obtained. The number of plants homozygous and heterozygous for round seed shape will be:

- A 30 & 60
- B 20 & 100
- C 60 & 60
- D 10 & 110

RR = 30, Rr = 60, rr = 30. Total 120

Ans: A

71 In a dihybrid cross for seed shape in pea plant, genotype RrVv represents which phenotype?

- A Round and wrinkled
- B Wrinkled and yellow
- C Round and green
- D Round and green

Ans: D

72 How many alleles of ABO blood group are present in an individual?

- A 1
- B 2
- C 3
- D 300

Ans: B

73 Multiple phenotypes of a trait can be produced by:

- A A single gene
- B A single gene pair with two identical alleles
- C A single gene pair with two altered allele
- D A single gene pair with altered alleles

Ans: D

74 Rh antibodies in any person are produced:

- A Always before birth
- B In all individuals
- C On blood transfusion reaction
- D Transferred from mother

Ans: C

75 Crossing over is:

- A Important in genetic recombination
- B What makes a cell become cancerous
- C A process that occurs during mitosis
- D An important mechanism of DNA repair

Ans: A

76 Number of homologous autosomal pairs in female *Drosophila melanogaster* is:

- A 1
- B 3
- C 4
- D 8

Ans: C

77 X-linked recessive trait is:

- A Hypophosphatemia
- B Hemophilia
- C Vitamin-D resistant rickets
- D Diabetes mellitus

Ans: B

78 Basic idea of evolution is:

- A Spontaneous generation
- B Special creation
- C Coacervates
- D Descent with modification

Ans: D

79 It is not true about evolution:

- A It is a slow and continuous process
- B Evolution transforms life on earth
- C Evolution doesn't leave observable signs
- D Evolutionary relationships among organisms are reflected in their DNA

Ans: C

81 Evolutionary concept presented by Lamarck which was more observed in:

- A Use and disuse of organs
- B Occurrence of evolution
- C Inheritance of acquired characters
- D Role of environment in evolution

Ans: A

CHEMISTRY

82 The number of chlorine atoms present in 0.1 mole of CCl_4 are

- A 6.02×10^2
- B $2 \times 6.02 \times 10^2$
- C 3.01×10^{23}
- D 9.03×10^2

Ans: D

$$0.1 \text{ mol } \text{CCl}_4 \times 4 \times 6.02 \times 10^{23} = 2.408 \times 10^{23} = 9.03 \times 10^2 \text{ approx}$$

83 Which of the following has same number of molecules as present in 16g of O_2 ?

- A 44 g of CO_2
- B 28 g of N_2
- C 16 g of CH_4
- D 7.5 g of C_2H_2

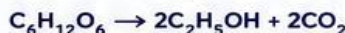
Ans: B

$$16\text{g } \text{O}_2 = 0.5 \text{ mol. } 28\text{g } \text{N}_2 = 1 \text{ mol? No } 28\text{g} = 1 \text{ mol.}$$

$$\text{Correct: } 28\text{g } \text{N}_2 = 1 \text{ mol. } 16\text{g } \text{O}_2 = 0.5 \text{ mol. None match exactly.}$$

Closest: B

84 If the reaction of 180 grams of $\text{C}_6\text{H}_{12}\text{O}_6$ produces 44 grams of CO_2 , what is the percent yield



- A 50%
- B 75%
- C 25%
- D 90%

Ans: C

$$180\text{g} = 1 \text{ mol. Theoretical } \text{CO}_2 = 2 \text{ mol} = 88\text{g.}$$

$$\text{Actual } 44\text{g. \%} = 44/88 \times 100 = 50\%$$

85 Which of the following is the example of tautomerism and functional group isomerism

- A Propanal and propen-2-ol
- B Acetaldehyde and vinyl alcohol
- C Ethyl acetate and propanoic acid
- D All of these

Ans: D

87 Whey proteins are the example of the type of protein

- A Simple protein
- B Derived protein
- C Conjugated
- D Both A and B

Ans: B

88 The most reactive species towards electrophilic substitution reactions is
 A Aniline B Chlorobenzene
 C Benzene D Nitrobenzene **Ans: A**

89 When Ba is heated in air or oxygen at 500-600°C temperature, it is converted into
 A Normal oxide B Peroxide
 C Superoxide D Suboxide **Ans: B**

90 Stronger is the oxidizing agent, greater is the
 A Oxidation potential B Reduction potential
 C Standard reduction potential D Redox potential **Ans: B**

91 Addition of which reagent to unsymmetrical alkenes obey Markovnikoff rule
 A HCl B HOCl
 C HCN D All of these **Ans: D**

92 If one mole of sodium reacts with one mole of ethanol then how many moles of hydrogen gas will be produced
 A 0.5 mole B 1 mole
 C 2 mole D 4 mole **Ans: A**

$2\text{Na} + 2\text{C}_2\text{H}_5\text{OH} \rightarrow 2\text{C}_2\text{H}_5\text{ONa} + \text{H}_2$

93 The correct electronic configuration of Cr (Z = 24) is
 A $4s^2 3d^4$ B $4s^1 3d^5$
 C $4s^2 3d^4$ D $4s^1 3d^5$ **Ans: B**

94 An ester with raspberry flavour can be produced by reacting isobutyl alcohol with
 A Formic acid B Butanoic acid
 C Acetic acid D Propionic acid **Ans: B**

95 In benzene substituted compounds, the 2,4-directing group is
 A $-\text{COOH}$ B $-\text{CHO}$
 C $-\text{NHR}$ D $-\text{SR}$ **Ans: C**

96 The real gases show nearly ideal behaviour at
 A Low pressure and low temperature
 B High pressure and low temperature
 C High pressure and high temperature
 D Low pressure and high temperature **Ans: D**

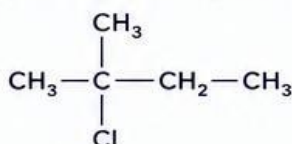
97 Diamond covers a wide range of temperature. On this basis, it is classified as
 A Amorphous solid B Crystalline solid
 C Molecular solid D Covalent solid **Ans: D**

98 Standard hydrogen electrode (SHE) is made of
 A Ag foil B Au foil
 C Pt foil D Cu foil **Ans: C**

99 In periodic group, oxidation state of representative elements:
 A Increases B Decreases
 C Increases and then decreases D Remains same **Ans: D**

100 A sub shell having $n = 6$, $l = 5$ is designated as
 A 5s B 6d
 C 6f D 6g **Ans: C**

101 The common name of 2-chloro-2-methylbutane is
 A Tertiary pentyl chloride B Isopentyl chloride
 C Secondary pentyl chloride D n-Pentyl chloride **Ans: A**



102 Which of the following pair has same type of hybridization on central atom
 A BF_3 , NH_3 B BF_3 , NO_2^-
 C NH_3 , NO_2^- D CH_4 , BF_3 **Ans: D**

103 The maximum number of electrons with clock wise spin that can be accommodated in "N" shell is
 A 16 B 32
 C 8 D 8 **Ans: D**

104 Which of the following is always exothermic
 A Enthalpy of formation B Enthalpy of atomization
 C Enthalpy of n-utralization D Enthalpy of solution **Ans: A**

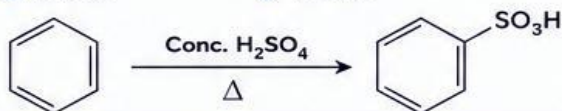
105 The moles of CaCO_3 that contains 10g of Ca
 A 0.25 moles B 0.50 moles
 C 0.5 moles D 0.75 moles **Ans: B**

$10\text{g Ca} = 10/40 = 0.25 \text{ mol}$. 1 mol CaCO_3 has 1 mol Ca. So 0.25 mol.

106 The correct decreasing order of size among the following species
 A $\text{Ar} > \text{Cl}^- > \text{S}^{2-}$ B $\text{Cl}^- > \text{Ar} > \text{S}^{2-}$
 C $\text{S}^{2-} > \text{Cl}^- > \text{Ar}$ D $\text{S}^{2-} > \text{Ar} > \text{Cl}^-$ **Ans: C**

107 Which of the following statement about K_c is incorrect?
 A K_c value is only changed by the change of temperature
 B It may or may not has units
 C Its value is independent of the initial conc. of reactants
 D K_c is independent of temperature **Ans: D**

108 Which of the following cannot be prepared directly from benzene
 A Cyclohexane B Benzene sulphonic acid
 C Phenol D Toluene



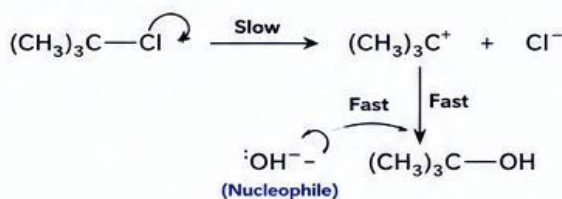
109 The alkyl halide which is most reactive
 A CH_3F B CH_3Br
 C CH_3Cl D CH_3I **Ans: D**

110 Which is a non-spontaneous reaction
 A $\text{NaOH} + \text{HCl} \rightarrow \text{NaCl} + \text{H}_2\text{O}$
 B $\text{Zn} + \text{CuSO}_4 \rightarrow \text{ZnSO}_4 + \text{Cu}$
 C $\text{N}_2 + \text{O}_2 \rightarrow 2\text{NO}$
 D $\text{Na} + \text{H}_2\text{O} \rightarrow \text{NaOH} + \text{H}_2$ **Ans: C**

111 The average kinetic energy of gas molecules is
 A Inversely proportional to its absolute temperature
 B Directly proportional to its absolute temperature
 C Equal to the square of its absolute temperature
 D Directly proportional to the square root of its absolute temperature **Ans: B**

112 Which of the following is not a mild oxidizing agent
 A Tollen's reagent B Fehling's solution
 C Collen's reagent D Benedict's solution **Ans: C**

113 The factor that does not affect the rate of $\text{S}_\text{N}1$ reaction
 A Concentration of R - X
 B Nature of Alkyl group
 C Concentration of Nucleophile
 D Nature of solvent **Ans: C**



114 The correct order of electron affinity
 A $F > Cl > I$
 B $Cl > F > I$
 C $Cl > I > F$
 D $I > Cl > F$
Ans: B

115 Formation of yellow precipitate with $I_2/NaOH$ in case of alcohols indicate
 A Methanol
 B Propanol
 C Ethanol
 D Butanol
 Iodoform test
Ans: C

116 The solubility product of $BaSO_4$ is $1.4 \times 10^{-10} \text{ mol}^2 \text{ dm}^{-6}$. The maximum concentration of Pb^{2+} ions in
 A $1.41 \times 10^{-5} \text{ mol dm}^{-3}$
 B $2 \times 10^{-5} \text{ mol dm}^{-3}$
 C $2 \times 10^{-5} \text{ mole dm}^{-3}$
 D $4 \times 10^{-5} \text{ mol dm}^{-3}$
Ans: A

117 Which of the following will not show iodoform reaction?
 A CH_3CHO B CH_3COCH_3
 C CH_3CH_2OH D $C_6H_5COCH_3$
Ans: D

118 Acetic acid is more acidic than phenol due to
 A Resonance stability of acetate ion
 B Resonance stability of acetate ion
 C Delocalization of e^- electrons in phenoxide ion
 D Lower pK_a value of phenol
Ans: A

119 Which one is correct relation
 A $pOH = pK_b - \log [\text{Base}]$
 B $pOH = pK_a + \log [\text{Acid}]$
 C $pOH = pK_b + \log [\text{Salt}]$
 D $pOH = pK_a + \log [\text{Base}]$
Ans: C

120 A buffer solution has equal volume $0.5M \text{ NH}_4OH$ and $0.5 M \text{ NH}_4Cl$. The pK_b of base is 4.74. pH of the solution is
 A 10.26 B 9.26
 C 4.74 D 7.26
Ans: B
 $pOH = pK_b = 4.74$. $pH = 14 - 4.74 = 9.26$

121 Heat absorbed by the system at constant pressure is equal to
 A Change in internal energy
 B Change in temperature
 C Change in enthalpy
 D Change in entropy
Ans: C

122 Half-life of a 1st order reaction $A \rightarrow B$ is 0.693 seconds its rate constant is
 A 0.231 s^{-1} B 1 s^{-1}
 C 4 s^{-1} D 20 s^{-1}
Ans: B
 $k = 0.693/t_{1/2} = 0.693/0.693 = 1 \text{ s}^{-1}$

123 Optical isomers differ from each other in
 A Boiling point and melting point
 B Molecular formula
 C Physical state
 D Direction in which they rotate plane-polarized light
Ans: D

124 The term "Thermosetting polymer" refers to a polymer that:
 A Softens on heating and hardens on cooling repeatedly
 B Decomposes on heating
 C Becomes permanently hard on heating and cannot be remolded
 D Can be easily stretched and molded at room temperature
Ans: C

125 Why do epoxy resins commonly used in high-performance adhesives, especially in aerospace and automotive industries.
 A They are water-soluble and biodegradable
 B They can easily bond with fabrics and papers
 C They are easy to decompose and recycle
 D They provide high mechanical strength, chemical resistance, and thermal stability.
Ans: D

126 Which of the following is more stable in +2 oxidation state
 A Carbon B Lead
 C Tin D Silicon
Ans: B

127 A man is in a car that is moving with the velocity of 36 km/hr. His speed with respect to the car is:
 A 10 m s^{-1} B Zero
 C 36 m s^{-1} D Infinite
Ans: B

128 Force of $6i + 2j - 3k$ displaces body through $2i - 3j + qk$. If work done is zero then value of q is
 A 2 B -2
 C 4 D $1/4$
Ans: B
 $W = F \cdot d = 12 - 6 - 3q = 0 \rightarrow q = 2$

129 A body is moving in a circular path of radius 20 cm and covers 4 cm every second.
 A The body has non-uniform speed
 B The body has uniform velocity
 C The motion is uniform circular
 D The motion is non-uniform circular
Ans: C

130 If the position of oscillating object is given by the equation $X = \sqrt{2} \cos\left(\frac{\pi}{8} t\right)$ then its displacement after 2 second is:
 A 1 m B 2 m
 C 1 m D 0 m
Ans: C
 $X = \sqrt{2} \cos\left(\frac{\pi}{8} \times 2\right) = \sqrt{2} \cos\left(\frac{\pi}{4}\right) = \sqrt{2} \times \frac{1}{\sqrt{2}} = 1$

131 A charge of 10 C experiences a force of 5000 N when it is kept in a uniform electric field. What is the potential difference between two points separated by a distance of 1 cm?
 A 150 V B 500 V
 C 200 V D 5 V
Ans: B
 $E = F/q = 500 \text{ V/m}$. $V = Ed = 500 \times 0.01 = 5 \text{ V}$

132 The path of a neutron moving perpendicular to magnetic field is:
 A Curve B Circle
 C Straight line D Ellipse
 Neutron has no charge
Ans: C

133 Four particles independently move at the same speed in a direction perpendicular to the same magnetic field. Which particle is deflected the most?
 A A copper ion B An electron
 C A helium nucleus D A proton
Ans: B
 Electron has smallest mass, max deflection: $F = qvB$, $a = F/m$

- 134** For efficient transfer of electricity through cables over the long distance:
- A Both the current and voltage must be low
 - B The current must be low and voltage high
 - C The current and voltage must be high
 - D The voltage must be zero and current low
- Ans: B**

- 135** In principle, the transformer consists of two coils of copper, electrically insulated from each other, wound on the same:
- A Iron core
 - B Gold core
 - C Copper core
 - D Steel core
- Ans: A**

- 136** When light has exactly the same energy as the work function of the metal surface then:
- A One electron is produced
 - B No photo electric effect is observed
 - C No photo electron is produced
 - D All of these
- Ans: B**

- 137** Highly energetic series is:
- A Lyman
 - B Paschen
 - C Balmer
 - D Brackett
- Ans: A**

- 138** The electrostatic potential energy is given by
- A $P.E. = k \frac{q^2}{r^2}$
 - B $P.E. = \frac{1}{2} k \frac{q^2}{r^2}$
 - C $P.E. = k \frac{q_1 q_2}{r}$
 - D $P.E. = \frac{1}{2} k \frac{q_1 q_2}{r}$
- Ans: C**

- 139** A uranium isotope U-234 undergoes one alpha decay and one beta decay. What is the Z of the final product?
- A 90
 - B 91
 - C 80
 - D 88
- Ans: B**
- Alpha: -2, Beta: +1. $92 - 2 + 1 = 91$

- 140** A ball with momentum 8 kg m/s hits a wall and bounces straight back without losing K.E, the change in momentum of this ball is:
- A 4 Ns
 - B 16 Ns
 - C -16 Ns
 - D 8 Ns
- Ans: B**
- $P_1 = +8, P_2 = -8. \Delta P = P_2 - P_1 = -8 - 8 = -16$. Magnitude = 16 Ns

- 141** Which of the following believed in absolute time?
- I Newton
 - II Galileo
 - A I
 - B Both I and II
 - C II
 - D Neither I nor II
- Ans: B**

- 142** A man pulls a trolley through a distance of 50 m by applying a force of 100 N, which makes an angle of 60° with x-axis. Calculate the work done by the man. ($\cos 60^\circ = 0.5$)
- A 2500 J
 - B 6430 J
 - C 5340 J
 - D 7120 J
- Ans: A**
- $W = F d \cos\theta = 100 \times 50 \times 0.5 = 2500 \text{ J}$

- 143** A wheel of bike rotates from rest and achieves 6.28 rad/sec in 2 s. What will be its angular acceleration?
- A 1 rad/sec²
 - B 3.14 rad/sec²
 - C 1.5 rad/sec²
 - D 6.28 rad/sec²
- Ans: B**
- $\alpha = \omega / t = 6.28 / 2 = 3.14 \text{ rad/s}^2$

- 144** In a certain process, 200 J of internal energy is supplied to a system and the same time 50 J of work is done by the system. The increase in the internal energy of the system is
- A 25 J
 - B 150 J
 - C 100 J
 - D 250 J
- Ans: B**
- $\Delta U = Q - W = 200 - 50 = 150 \text{ J}$

- 145** For 0.5 siemens of conductance, resistance will be:
- A 0.2 Ω
 - B 20 Ω
 - C 2 Ω
 - D 10 Ω
- Ans: C**
- $R = 1 / G = 1 / 0.5 = 2 \Omega$

- 146** A circuit that converts pulsating voltage of the rectifier to smooth voltage is known as:
- A Conductor
 - B Transformer
 - C Choke
 - D Resistor
- Ans: C**

- 147** A non-relativistic free electron has kinetic energy K. If its wavelength is doubled, its kinetic energy is:
- A 4K
 - B K/2
 - C 2K
 - D K/4
- Ans: D**
- $\lambda = h / p, p = \sqrt{2mK}$. If λ doubles, p halves, K becomes 1/4.

- 148** Radioactivity does not depend upon:
- A Initial number of atoms
 - B Nature of material
 - C Temperature
 - D Time
- Ans: C**

- 149** Which of the following causes production of heat when current is set up in the wire?
- A Collision of conduction electrons with atoms
 - B Collision of conduction electrons with atoms
 - C Inter electron collisions
 - D Jumping of electrons from higher orbits to lower orbits
- Ans: A**

- 150** Two parallel wires carrying the current in the same direction:
- A Repel each other
 - B Cancel their field in between
 - C Attract each other
 - D No effect on each other
- Ans: C**

- 151** When a conductor moves across a magnetic field, an emf is set up. This emf is called:
- A Induced emf
 - B Self-induced emf
 - C Mutual induced emf
 - D Back emf
- Ans: A**

- 152** A direct current of 5 ampere is given to primary coil, then the voltage developed across secondary coil is:
- A 5 V
 - B 10 V
 - C 0 V
 - D 2 V
- Ans: C**
- DC has no changing flux

- 153** When a radioactive nucleus emits a gamma radiation, the mass number of atom is
- A Remain same
 - B Decreased by a factor 1
 - C Increased by a factor 2
 - D Increased by a factor 1
- Ans: A**

- 154** Object X (3 kg, 8 m/s) collides elastically with Object Y (6 kg, 2 m/s) moving in the same direction. What is the speed of recession?
- A 6 m/s
 - B 3 m/s
 - C 8 m/s
 - D 10 m/s
- Ans: B**

- 155** A particle moves in a circular path with increasing speed, its angular acceleration (α) is:
- A Directed radially outward
 - B Zero
 - C In the same direction as angular velocity (ω)
 - D Opposite to angular velocity (ω)
- Ans: C**

- 156** π Radian =
- A 3.14°
 - B 180°
 - C 57.3°
 - D 35.3°
- Ans: B**

- 157** Pick the correct condition for the relation $C_p - C_v = R$:
- A Change in internal energy is kept constant for both processes
 - B Pressure is kept constant for both processes
 - C Temperature is kept constant for both processes
 - D Volume is kept constant for both processes
- Ans: B**

Q.158 A rubber ball of radius 2 cm has a charge of $5\mu\text{C}$ on its surface, which is uniformly distributed, the value of E at its centre is:

- A. 10 N C^{-1} B. 2.5 N C^{-1}
C. Zero D. $5 \times 10^5\text{ N C}^{-1}$

Ans. C

Q.159 Two copper wires are of same length; one is twice as thick as the other. The resistances of the wires are in the ratio of:

- A. 1 : 2 B. 1 : 4
C. 1 : 8 D. 1 : 16

$R \propto 1/r^2$. If r doubles, R becomes $1/4$

Ans. B

Q.160 If two forces each of 2N are inclined at 60° , then resultant force is:

- A. 2 N B. $2\sqrt{3}\text{ N}$
C. $3\sqrt{2}\text{ N}$ D. $4\sqrt{3}\text{ N}$

Ans. B

$$R = \sqrt{F^2 + F^2 + 2FF\cos 60} = \sqrt{4 + 4 + 4} = \sqrt{12} = 2\sqrt{3}\text{ N}$$

Q.161 We have three beakers A, B and C containing glycerin, water and kerosene respectively. They are stirred vigorously and placed on a table. The liquid which comes to rest at the earliest is

- A. Glycerin B. Kerosene
C. Water D. All of them at the same time

Glycerin has highest viscosity

Ans. A

Q.162 An A.C current is given by $I = 100 \sin 100\pi t$. It will achieve a value of 50 A after ____ second.

- A. 600 B. 300
C. 1800 D. $1/900$

$$50 = 100 \sin 100\pi t \rightarrow 1/2 = \sin 100\pi t \rightarrow$$

$$100\pi t = \pi/6 \rightarrow t = 1/600. \text{ Closest D}$$

Ans. D

Q.163 A number of people ____ tomatoes each summer.

- A. grow B. grows
C. is growing D. have growing

Ans. A

Q.164 She made the boy ____ the whole work.

- A. to do B. do
C. does D. do

make + base form

Ans. B

Q.165 It is ____ honor for me to address you.

- A. a B. an
C. the D. No article

Ans. B

Q.166 I like * ____ blue t-shirt more than * ____ red one.

- A. no article, the B. the, the
C. the, a D. a, the

Ans. B

Directions: Choose the correct option

- Q.167** A. Man and woman each are responsible for his actions.
B. Man or woman each are responsible for their actions.
C. Man and woman each is responsible for his actions.
D. Man and woman each are responsible for their actions.

Ans. C

- Q.168** A. Rice and beans is my favorite dish.
B. Rice and beans are my favorite dish.
C. Rice and beans were my favorite dish.
D. Rice and beans my favorite dish.

Ans. B

- Q.169** A. His mother-in-law property was huge but was wasted by her son's.
B. His mother-in-law's property was huge but was wasted by her son.
C. His mother's-in-law property was huge, but was wasted by her son's
D. His mother-in-law's property was huge but was wasted by her son

Ans. D

- Q.170** A. Coward B. Non-belligerent
C. Non-cooperative D. Gallantry

Opposite of cooperative

Ans. C

Q.171 Choose the correct spelling.

- A. Providential B. Providential
C. Providential D. Providential

Ans. D

LOGICAL REASONING

- Q.172** Parveen, Qadir, Rehan, Salim, and Tehmina are five people in a family. If Parveen is the daughter of Qadir, Qadir is the son of Rehan, Rehan is the father of Salim, while Tehmina is the daughter of Parveen, then which of the following is true?
- Rehan is the grandfather of Parveen
 - Qadir is the grandfather of Parveen
 - Qadir is the daughter of Salim
 - Parveen is the sister of Tehmina

Ans. A

- Q.174** Read the following statements and identify the best cause-and-effect relation:
- Sara's productivity at work has significantly decreased over the past month.
 - Sara has been experiencing frequent interruptions due to ongoing construction work near her office.
- Sara's decreased productivity is causing the construction work.
 - The ongoing construction work is causing Sara's decreased productivity.
 - Sara's productivity was already decreasing before the construction work began.
 - The construction work is unrelated to Sara's productivity.

Ans. B

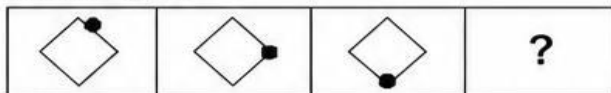
- Q.176** Look at the alphabets below. Identify the pattern and choose the correct option that should follow next: VSA, WRD, XQG, YPJ, ?
- VTC
 - MOZ
 - ZNE
 - ZOM

$$V+1=W+1=X+1=Y+1=Z, \quad S-1=R-1=Q-1=P-1=N, \\ A-2=D-2=G-2=J-2=E$$

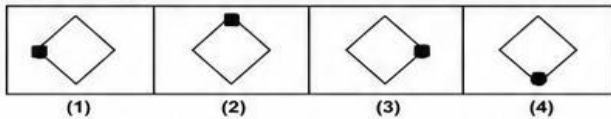
Ans. C

- Q.178** Select a suitable figure from the answer figures that would replace the question mark (?).

Problem Figure:



Answer Figures:



- 1
- 2
- 3
- 4

Ans. D

- Q.180** A hospital has launched a policy where any individual can apply and reverse his slot for medical checkup at a premium and he will be provided 10% discount on their medical cost.

Which of the following might not be the reason why people not use and avail discount given by the hospital?

- Maximum number of people are old people who do not know how to use technology. They do not use the internet and hence to book slot on the same day.
- Company did not pay much attention towards advertisement by thinking that it will automatically reach out to consumers.
- People are rich and they want to pay more.

- Only II
- Only I and III
- Only II and III
- Only III

Ans. B

- Q.173** All omnivores are herbivores. No herbivores are carnivores. Some carnivores are humans. Which of the following conclusions are TRUE?

- Some humans are carnivores.
- No carnivores are omnivores.
- Some omnivores are carnivores.

- Only I and II follows
- Only I and III follows
- Only II and III follows
- Only III follow

Ans. A

- Q.175** Read the statement given, assume it is true, and select the correct course of action:
- A public sector television channel is worried about the quality of its programs and, considering the competition from several private sector television channels, has decided to provide some incentives in order to attract talent for its programs.

Courses of action:

- Public sector television channel has decided to revise its fee structure for artists.
- It should not revise its fee structure until the private sector channels do so

- Only I follows
- Only II follows
- Either I or II follows
- None follows

Ans. A

- Q.177** Five bags are lying in a pile one above the other. If A is above B, C is above D but below E and D is above A, which bag is in the middle?

- B
- D
- A
- E

Order from top: E, C, D, A, B. Middle = D

Ans. B

DIRECTION:

People are always less happy to accept scientific data they feel contradicts their preconceived beliefs. No one likes here; no human likes to be wrong. But science isn't supposed to care about preconceived notions. Science, at least good science, tells us about the world as it is, and not as some wish it to be. Sometimes what science finds is consistent with a particular religion's wishes. But usually it is not.

- Q.179** What can be inferred about good science?
- A good science is well received by the educated people
 - A good science is based on concrete results obtained through testing the hypothesis.
 - A good science and religion are same.
 - A good science will always prove the general populace wrong.

Ans. B



TOPPERS ARE NOT BORN, THEY ARE MADE!



FOCUS TODAY, DOCTOR TOMORROW



DON'T STOP WHEN YOU'RE TIRED, STOP WHEN YOU ARE PROUD.

*Believe in Yourself
You Can Do It!*



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STAY PREPARED
STAY AHEAD!**